

ORIGINAL CONTRIBUTION

Tenecteplase Efficacy in Patients With Distal Occlusions: Secondary Analysis of the TASTE Randomized Clinical Trial

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BACKGROUND: Tenecteplase has been previously evaluated in large- and medium-sized vessel occlusion subgroups, but its effectiveness in more distal occlusions, particularly those involving the distal middle cerebral artery branches or the anterior cerebral artery and posterior cerebral artery territories, and across varying ischemic core and perfusion profiles, remains uncertain.

METHODS: We performed a secondary analysis of TASTE (Tenecteplase Versus Alteplase for Stroke Thrombolysis Evaluation), a randomized clinical trial comparing tenecteplase and alteplase in patients presenting within 4.5 hours of symptom onset with perfusion imaging-confirmed stroke and evidence of target mismatch (penumbra/core ratio >1.8 and an absolute difference >15 mL). The primary outcome was the modified Rankin Scale (mRS) score of 0 to 1 at 90 days. We compared the effect of tenecteplase versus alteplase in subgroups based on occlusion site (proximal M2, distal M2, M3 and beyond, anterior cerebral artery, and posterior cerebral artery), ischemic core, core growth rate, and penumbra. The treatment effect of tenecteplase and alteplase was compared stratifying by the subgroup of interest, adjusting for age, baseline National Institutes of Health Stroke Scale score, and premorbid mRS score in modified Poisson regression models.

RESULTS: Of the 680 patients enrolled, 492 were included in the primary analysis (median age, 73 [interquartile range, 63–82] years; male sex: 306/492 [62%]). Two hundred forty-two (49%) received tenecteplase, and 250 (51%) received alteplase. Tenecteplase was associated with a higher proportion of mRS score of 0 to 1 with distal (M3 and beyond) occlusions (tenecteplase: 62/81 [77%] versus alteplase: 59/93 [63%]; adjusted risk ratio, 1.23 [95% CI, 1.04–1.46]). Numerically higher rates of mRS score of 0 to 1 were observed with distal M2 (tenecteplase: 30/46 [65%] versus alteplase 28/48 [58%]; adjusted risk ratio, 1.14 [95% CI, 0.84–1.54]) and anterior cerebral artery occlusions (tenecteplase: 12/21 [57%] versus alteplase 5/13 [38%]; adjusted risk ratio, 1.46 [95% CI, 0.71–3.00]). No treatment differences were seen for proximal M2 or posterior cerebral artery occlusions ($P_{\text{interaction}}$ across all occlusion sites: 0.89). Across ischemic core, penumbra, and core growth rate subgroups, no difference in treatment effect was observed.

CONCLUSIONS: Patients with distal middle cerebral artery occlusions who are treated with tenecteplase are more likely to achieve an mRS score of 0 to 1 at 90 days than those treated with alteplase.

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GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: computed tomography angiography ■ middle cerebral artery ■ perfusion ■ stroke ■ tenecteplase

Nonstandard Abbreviations and Acronyms

ACA	anterior cerebral artery
ACT	Alteplase Compared to Tenecteplase in Patients With Acute Ischemic Stroke
CT	computed tomography
CTP	computed tomography perfusion
EXTEND-IA TNK	Extending the Time for Thrombolysis in Emergency Neurological Deficits-Intra-Arterial Using Intravenous Tenecteplase
MCA	middle cerebral artery
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
PCA	posterior cerebral artery
TASTE	Tenecteplase Versus Alteplase for Stroke Thrombolysis Evaluation

Several key landmark randomized controlled trials across Canada, Australia, the European Union, the United Kingdom, and China have examined the safety and efficacy of tenecteplase, demonstrating noninferiority compared with intravenous alteplase in the <4.5-hour time window.^{1–5} These trials assessed broad and heterogeneous patient populations, enhancing the generalizability of tenecteplase use for acute stroke management. Among these studies, the use of vascular imaging for enrollment was not uniformly mandated, and trials used this imaging modality to varying degrees. Tenecteplase efficacy has been previously evaluated in patients presenting with large or medium vessel occlusions via the EXTEND-IA TNK trial (Extending the Time for Thrombolysis in Emergency Neurological Deficits-Intra-Arterial Using Intravenous Tenecteplase) and ACT trial (Alteplase Compared to Tenecteplase in Patients With Acute Ischemic Stroke).^{6–9} However, the effectiveness of tenecteplase in patients with more distal vessel occlusions, such as those in the distal branches of the middle cerebral artery (MCA) or in the anterior or posterior cerebral artery territories (anterior cerebral artery [ACA] or posterior cerebral artery [PCA]), remains largely underinvestigated.

In addition to its ability to delineate ischemic core and penumbral tissue, computed tomography perfusion (CTP) offers the advantage of detecting distal ischemic lesions that may not be visible on conventional computed tomography (CT) angiography. The TASTE randomized clinical trial⁴ (Tenecteplase Versus Alteplase for Stroke Thrombolysis Evaluation) was the only phase 3 study to compare tenecteplase with alteplase in patients presenting with imaging-confirmed stroke via CTP and was, thereby, able to objectively identify distal occlusions of the MCA, ACA, and PCA. This secondary imaging analysis of TASTE was conducted to investigate the performance of tenecteplase across a range of occlusion sites and penumbral profiles.

METHODS



Data Availability Statement

The authors declare that all data and methodological details are available within the article and the [Supplemental Material](#). Access to the TASTE data set can be obtained from the senior investigator upon reasonable request.

Study Population

This was a post hoc secondary analysis of the TASTE trial (Unique identifier: ACTRN12613000243718). TASTE was a multicenter, open-label randomized clinical trial enrolling adults aged ≥18 years who presented within 4.5 hours of stroke onset or last known well to 35 stroke centers across Australia, Belgium, Canada, Finland, New Zealand, Spain, Taiwan, and the United Kingdom.⁴ The trial protocol was approved by the local institutional review board at all participating sites before recruitment began, and enrollment took place between March 2014 and October 2023. Patients or their legally authorized representatives provided written informed consent before participation, and in jurisdictions permitting deferred consent for emergency treatment, consent was obtained afterward. The trial protocol and statistical analysis plan for the main trial are available in the supplement of the main trial publication.⁴ This analysis followed the Consolidated Standards of Reporting Trials reporting guidelines.

Patients were eligible if they had imaging-confirmed acute ischemic stroke and evidence of salvageable tissue, requiring a CTP mismatch ratio >1.8 between the perfusion lesion (defined by delay time >3 seconds) and the ischemic core lesion (defined by relative cerebral blood flow <30%), an absolute mismatch difference >15 mL, ischemic core volume <70 mL, and severely hypoperfused tissue volume <100 mL. Patients were excluded if they had intracranial hemorrhage on initial imaging, premorbid disability with modified Rankin Scale

(mRS) score ≥ 2 , planned thrombectomy, or a contraindication to thrombolytics. For this analysis, the per-protocol population was utilized. The TASTE trial commenced during a period when endovascular therapy had not yet become established practice, resulting in the initial inclusion of a relatively small number of patients with large vessel occlusions (internal carotid artery, first segment [M1] segment of the MCA, or vertebrobasilar artery). As the current management of this population predominantly involves endovascular therapy, we also excluded these patients from our analysis.

Randomization, Blinding, and Interventions

Eligible patients were centrally randomized 1:1 to receive either intravenous tenecteplase or alteplase. Treatment allocation was open-label. All primary and secondary outcome assessments at 90 days were conducted centrally via a phone-based assessment by trial personnel who were blinded to treatment allocation. Upon randomization, patients received tenecteplase administered as a single bolus dose of 0.25-mg/kg body weight (maximum 25 mg) or alteplase at a dose 0.90-mg/kg body weight, with an initial bolus of 0.09 mg/kg followed by a 60-minute infusion of the remaining 0.81 mg/kg (maximum 90 mg). All other management is abided by local thrombolytic protocols for the standard of care in acute ischemic stroke.

Imaging Evaluation

Imaging assessments were performed by a central imaging laboratory consisting of vascular neurologists blinded to all clinical data and treatment assignment. All baseline CT scans (including computed tomography angiography and CTP), follow-up CT-magnetic resonance imaging scans done at 24 hours after admission, and subsequent scans completed during the patient's hospital admission were assessed by 2 blinded independent reviewers, with disagreement resolved by consensus or a third party. For this analysis, all baseline CTP scans were processed using the latest version of MISTar (Apollo Medical Imaging, Melbourne, Victoria, Australia). All scans were assessed for quality to ensure that patients met the original recruitment criteria. Baseline ischemic volume was defined using automated processing of relative cerebral blood flow of $<30\%$ on CTP or apparent diffusion coefficient $<620 \times 10^{-6} \text{ mm}^2/\text{s}$ on diffusion-weighted magnetic resonance imaging. The perfusion volume for each patient was defined using the delay time >3 seconds. Mismatch ratio was defined as the ratio between perfusion volume and ischemic core lesion volume. The mismatch volume (penumbra) was defined as the difference between the perfusion volume and ischemic core volume. To quantify the speed of infarction, we assessed core growth, calculated as core volume on baseline CTP (mL) divided by the time from stroke onset to initial CTP scan (hours). Our calculation of core growth rate assumed linear growth, and the core volume was zero just before stroke onset.^{10–12}

Vessel occlusion sites were based on the presence of the most proximal intracranial occlusion and were categorized as the internal carotid artery, the M1 of the MCA segment, the proximal M2 segment of the MCA, the distal M2 segment of the MCA, the M3 segment and beyond, the ACA, the PCA, and the vertebrobasilar artery. Distinguishing between a proximal

and distal M2 vessel was based on the midline point of the vessel. The contralateral side could be used as a reference, or if required (and available), postrecanalization information could be used to assess the underlying anatomy. For distal vessel occlusions that were difficult to appreciate on CT angiography, the topography of the acute perfusion lesion was used to assist classification.

Outcomes

Consistent with the main analysis, the primary outcome was the proportion of patients with no disability (mRS score, 0–1). Secondary outcomes included the proportion of patients achieving functional independence (mRS score, 0–2), the distribution of mRS score at 90 days, and the proportion of patients with severely disabling stroke or death at 90 days (mRS score, 5–6). Safety outcomes included symptomatic intracranial hemorrhage at 24-hour follow-up imaging. Symptomatic hemorrhage cases were adjudicated by a blinded committee of vascular neurologists and defined according to the European Cooperative Acute Stroke Study III criteria as any hemorrhage associated with neurological deterioration, marked by an increase in National Institutes of Health Stroke Scale (NIHSS) score of ≥ 4 points or death, where the hemorrhage was the primary cause of neurological worsening or death.¹³

Statistical Analysis

The Fisher exact test, the Mann-Whitney *U* test, and ANOVA were used as appropriate when evaluating baseline patient characteristics. All statistical tests were 2-tailed, with $P < 0.05$ defined as statistically significant. We evaluated subgroups defined by occlusion site, baseline ischemic core volume, core growth rate, and penumbral volume. The core growth rate was dichotomized using the mean and median values from the overall primary analysis population. Baseline ischemic core volume and penumbra were categorized into multiple strata, reflecting their distribution in the full cohort, to facilitate assessment of potential safety signals. We compared tenecteplase to alteplase for mRS score of 0 to 1 and mRS score of 0 to 2 while adjusting for age, baseline NIHSS score, and premorbid mRS score as prognostic variables in separate modified Poisson regression models with treatment effect reported as an adjusted risk ratio. An adjusted generalized odds ratio was used to report ordinal mRS score with adjustments for age, baseline NIHSS score, and premorbid mRS score, and treatment effect was reported as an adjusted common odds ratio. Firth logistic regression was used to assess mRS score of 5 to 6, adjusting for age, baseline NIHSS score, and premorbid mRS score, with treatment effect reported as an adjusted odds ratio. Because symptomatic intracranial hemorrhage was rare in the main trial, the treatment effect was assessed using an unadjusted risk difference. Hemorrhagic transformation subtypes were reported as absolute counts and proportions, without further comparative analysis. No imputation was performed for missing data. Patients with missing 90-day functional outcome, baseline CT or magnetic resonance perfusion imaging, or follow-up scans were excluded from analyses. These analyses were exploratory in nature, with no adjustments made for multiple comparisons. All analyses were conducted using SPSS (version 30.0) and R software (version 4.1.3).

RESULTS

Cohort Characteristics

Of the total 680 patients recruited into the study, 492 patients were included in the primary analysis, with 242 (49%) patients receiving tenecteplase and 250 (51%) receiving alteplase (reasons for study exclusion are outlined in Figure 1). Baseline characteristics stratified by treatment and by occlusion site are outlined in Tables 1 and 2, respectively. Between treatment groups, baseline features were well balanced, and ischemic core volumes and median mismatch ratios and volumes were similar between groups. The distribution of occlusion sites was similar between groups, with most cases involving MCA occlusions beyond the M1 (proximal M2: n=98 [19.9%]; distal M2: n=94 [19.1%]; M3 and beyond: n=174 [35.3%]; ACA: n=34 [6.9%]; and PCA: n=92 [18.7%]). Between vascular territories, age, sex, and workflow processes were well balanced. Relative to MCA-territory occlusions, PCA occlusions were associated with lower NIHSS scores, reduced ischemic core and mismatch volumes, and a higher mismatch ratio, reflecting a milder overall presentation. No major differences in clinical severity and baseline imaging features were observed between MCA and ACA occlusions.

Treatment Efficacy by Occlusion Type

Ninety-day mRS score distributions across MCA, ACA, and PCA territories, stratified by treatment, are presented in Figure 2. Across all MCA patients, tenecteplase (119/175 [68%]) yielded a higher proportion of mRS score of 0 to 1 compared with alteplase (116/191 [60%]) and demonstrated a significant shift toward better outcomes across the mRS score spectrum (adjusted common odds ratio, 1.34 [95% CI, 1.06–1.69]; Figure 2). No significant treatment interaction was observed across occlusion sites ($P_{\text{interaction}}=0.89$). Within occlusion subgroups, tenecteplase (77% [62/81]) was associated with a higher proportion of mRS score of 0 to 1 over alteplase (63% [59/93]) in patients with distal MCA (M3 and beyond) occlusions (adjusted risk ratio, 1.23 [95% CI, 1.04–1.46]). A higher proportion of patients achieving an mRS score of 0 to 1 with tenecteplase was also observed with distal M2 and ACA occlusions; however, these differences were not statistically significant. No differences in the rates of mRS score of 0 to 1 were noted for proximal M2 or PCA occlusions (Table 3). These results were reproducible across both a dichotomized mRS score of 0 to 2 analysis (Table S1) and an ordinal mRS score evaluation (Table S2). No differences

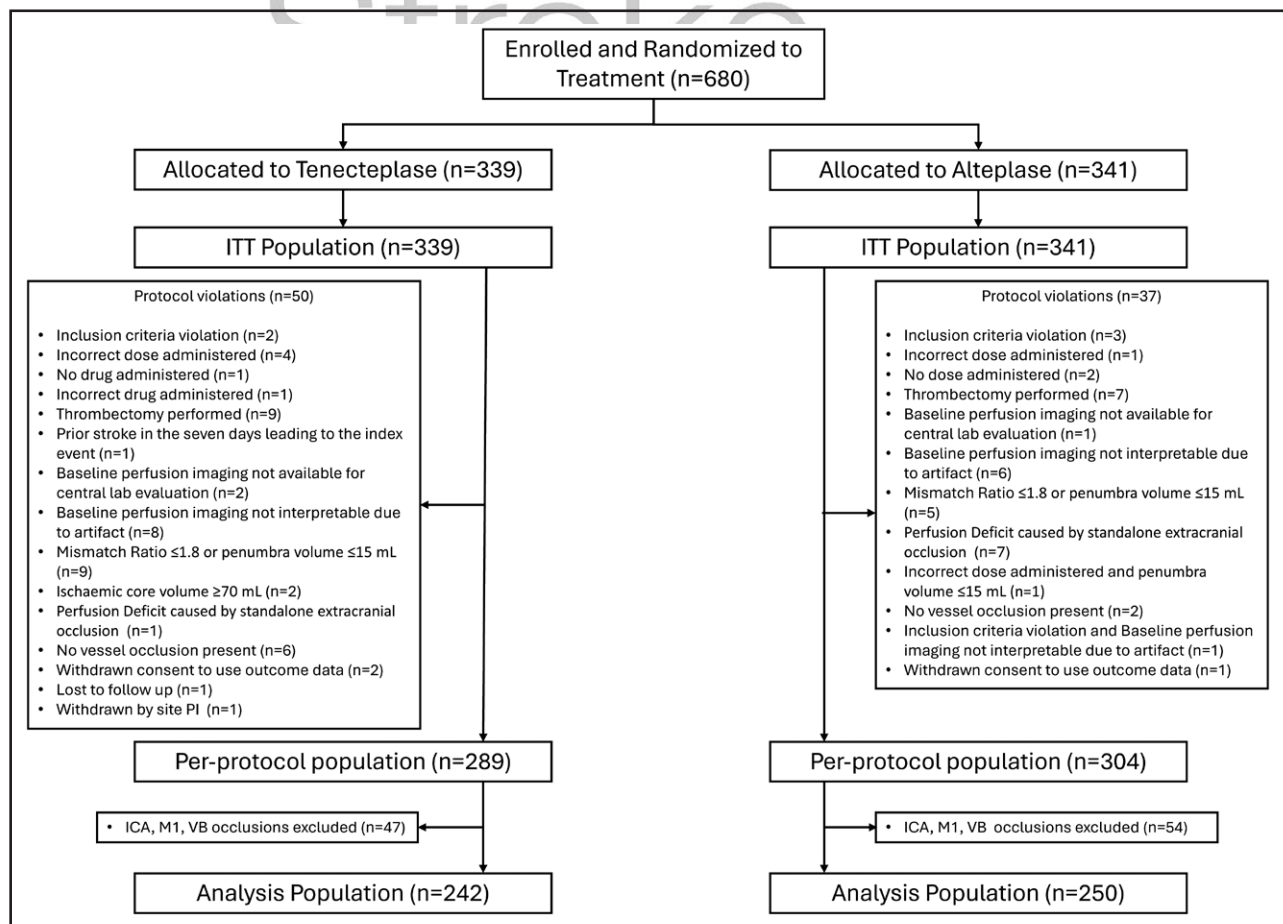


Figure 1. Consolidated Standards of Reporting Trials diagram.

Table 1. Baseline Patient Characteristics Stratified by Treatment (n=492)

	Tenecteplase (n=242)	Alteplase (n=250)
Age, y; median (IQR)	73 (62–81)	73 (63–82)
Male sex	148 (61%)	158 (63%)
NIHSS score, median (IQR)	7 (4–10)	7 (4–9)
Medical history		
Hypertension	154 (64%)	157/249 (63%)
Diabetes	46 (19%)	48/249 (10%)
Elevated cholesterol	100 (41%)	96/249 (39%)
Atrial fibrillation	43 (18%)	47 (19%)
Ischemic heart disease	46 (19%)	50/249 (20%)
Previous stroke	29 (12%)	27 (11%)
Workflow times, min; median (IQR)		
Stroke onset or last known well to hospital arrival	74 (55–105)	86 (58–127)
Stroke onset or last known well to thrombolytic administration	146 (118–188)*	156 (126–200)
Hospital arrival to thrombolytic administration	63 (47–86)*	64 (50–84)
Baseline imaging characteristics		
Perfusion CT, mL; median (IQR)		
Ischemic core volume†	7.0 (3.0–14.4)	5.9 (2.2–10.1)
Mismatch ratio§	5.1 (2.7–10.3)	5.0 (2.8–11.6)*
Mismatch volume (penumbra)	25.0 (15.2–35.4)	21.8 (14.0–35.9)
Core growth rate, mL/h; median (IQR)¶	2.9 (1.4–5.9)	2.2 (1.0–5.3)
Occlusion site		
MCA (second segment [M2], proximal)	48 (20%)	50 (20%)
MCA (second segment [M2], distal)	46 (19%)	48 (19%)
MCA (third segment [M3] and beyond distally)	81 (34%)	93 (37%)
Anterior cerebral artery	21 (9%)	13 (5%)
Posterior cerebral artery	46 (18%)	46 (19%)

Results are n (%) unless otherwise stated. CT indicates computed tomography; CTP, computed tomography perfusion; IQR, interquartile range; MCA, middle cerebral artery; and NIHSS, National Institutes of Health Stroke Scale.

*Missing 1 patient.

†Relative cerebral flow at a threshold of 30%.

‡Delay time >3 s.

§Ratio between total perfusion volume and ischemic core lesion volume.

||Defined as the difference between total perfusion volume and ischemic core lesion volume.

¶Defined as core volume on baseline CTP (mL) divided by the time from stroke onset to initial CTP scan (hours).

in mRS score of 5 to 6 (Table S3) or symptomatic intracranial hemorrhage (Table S4) were noted across any occlusion subgroup. While no symptomatic hemorrhages were observed with either ACA or PCA occlusions, an increased proportion of hemorrhagic transformations was observed among PCA occlusions (21/46, 46%) compared with MCA (67/366, 18%) or ACA (6/34, 17%) occlusions. Between treatment groups, no difference in hemorrhagic transformation subtypes was noted (Table S5).

Treatment Efficacy by Ischemic Core, Core Growth Rate, and Penumbra

Across 492 patients, ischemic core ranged from 0 mL (fifth percentile) to 33 mL (95th percentile). Based on this range, ischemic core was grouped into 10-mL strata

(<10, 10–20, 20–30, and >30 mL), encompassing the full distribution of ischemic core within the study population. Across ischemic core subgroups, no difference in treatment effect was observed for the primary (Table 3; $P_{\text{interaction}}=0.69$) or secondary outcomes (Tables S1 and S2). Rates of mRS score of 5 to 6 increased with larger ischemic core strata, from 4% (15/331) in patients with core <10 mL to 21% (13/63) in those with core >20 mL. However, no differences between treatment groups were observed (Table S3). Similarly, rates of symptomatic hemorrhage and all forms of hemorrhagic transformation were comparable between the groups (Tables S4 and S6).

The median and mean core growth rates across all patients were 2.7 (interquartile range, 1.0–5.6) mL/h and 4.4 (SD, 6.4) mL/h, respectively. As such, a threshold of 3 mL/h was selected. At this threshold, we did not

Table 2. Baseline Characteristics Stratified by Vascular Territory (n=492)

	MCA (n=366)	ACA (n=34)	PCA (n=92)
Age, y; median (IQR)	73 (63–82)	70 (63–82)	73 (60–80)
Male sex	220 (60%)	23 (67.6%)	62 (69%)
NIHSS score, median (IQR)	7 (4–10)	8 (5–13)	5 (3–8)
Workflow times, min; median (IQR)			
Stroke onset or last known well to hospital arrival	78 (56–119)	91 (61–141)	81 (55–108)
Stroke onset or last known well to thrombolytic administration	151 (120–199)	153 (126–218)	150 (120–188)
Hospital arrival to thrombolytic administration	64 (48–86)	63 (53–80)	63 (45–82)
Baseline imaging characteristics			
Occlusion subtype			
Proximal M2	98 (27%)		
Distal M2	94 (26%)		
M3 and beyond	174 (48%)		
A1/A2		23 (68%)	
Distal ACA		11 (32%)	
P1/P2			80 (87%)
Distal PCA			12 (13%)
Perfusion CT, mL; median (IQR)			
Ischemic core volume*	7.0 (3.0–13.0)	5.9 (2.5–11.2)	3.0 (1.1–7.6)
Mismatch ratio†	5.6 (3.6–9.5)	5.8 (4.2–10.5)	10.4 (3.9–25.1)
Mismatch volume (penumbra)‡	32.0 (21.0–46.1)	32.8 (26.0–42.8)	24.0 (17.5–36.3)
Core Growth Rate, mL/h; median (IQR)	3.2 (1.2–5.9)	1.9 (1.3–5.4)	1.4 (0.5–2.9)

Results are n (%) unless otherwise stated. ACA indicates anterior cerebral artery; CTP, computed tomography perfusion; IQR, interquartile range; MCA, middle cerebral artery; NIHSS, National Institutes of Health Stroke Scale; and PCA, posterior cerebral artery.

*Relative cerebral flow at a threshold of 30%.

†Delay time >3 s.

‡Ratio between total perfusion volume and ischemic core lesion volume.

§Defined as the difference between total perfusion volume and ischemic core lesion volume.

||Defined as core volume on baseline CTP (mL) divided by the time from stroke onset to initial CTP scan (hours).

observe a difference in treatment effect for both primary (Table 3) and secondary outcomes (Tables S1 through S3). No differences in symptomatic hemorrhage were noted among core growth rate subgroups (Table S4). The median and mean penumbra across all patients were 32 (interquartile range, 21–44) mL and 37 (SD, 23) mL, respectively. As such, strata based on the median and interquartile range were selected for further analysis. Improved rates of mRS score of 0 to 1 were observed in tenecteplase-treated patients with lower volumes of penumbra (<25 mL), but, in penumbra strata >25 mL, we did not observe a difference in treatment effect for both primary (Table 3) and secondary outcomes (Tables S1 through S3). No differences in symptomatic hemorrhage or hemorrhage subtypes were noted (Tables S4 and S7).

DISCUSSION

In a post hoc secondary analysis of the TASTE trial, tenecteplase demonstrated improved clinical outcomes compared with alteplase in patients with distal vessel occlusions and most clearly for distal occlusions of the MCA territory. Across all occlusion sites and perfusion profiles evaluated, rates of symptomatic ICH and

mortality were low, reinforcing a favorable safety profile of thrombolysis in this patient cohort. These observations add to a limited evidence base, as comparative data on tenecteplase versus alteplase in patients with distal vessel occlusions remain sparse.⁷

The use of perfusion in this study cohort allowed us a unique opportunity to objectively identify distal occlusions that would otherwise not be easily assessed by CT angiography alone. Most distal occlusions in this trial were identified by evaluating the topography of the acute perfusion lesion, underscoring the added diagnostic value of perfusion imaging in this context. While there remains a theoretical risk that artifact or noise could be misinterpreted as a true ischemic lesion, this was mitigated by requiring a minimum mismatch volume of 15 mL and a mismatch ratio >1.8 for enrollment into TASTE. In addition, our use of the per-protocol population focuses the analysis on patients who met all imaging and clinical criteria, providing a biologically coherent sample in which to evaluate any tenecteplase treatment effect.

The use of perfusion imaging also allowed us to assess tenecteplase against clearly quantified volumes of ischemic core and penumbra. No differences in treatment effect were noted across ischemic core strata,

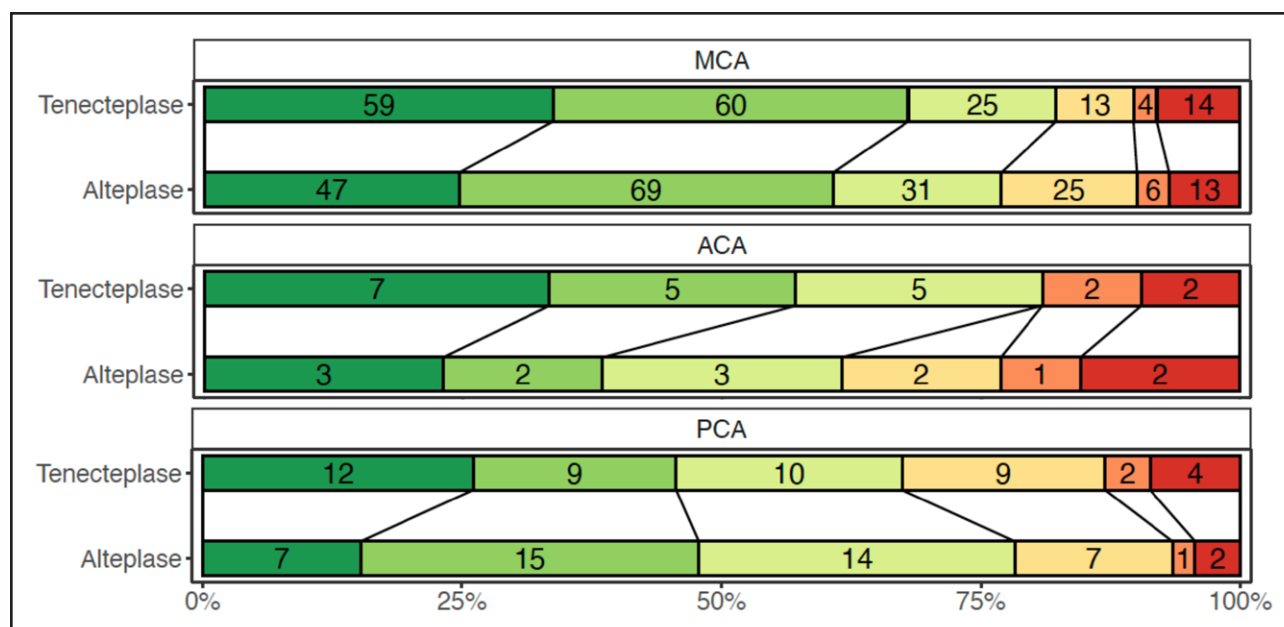


Figure 2. Distributions of modified Rankin Scale scores at 90 days stratified by occlusion site.

ACA indicates anterior cerebral artery; MCA, middle cerebral artery; and PCA, posterior cerebral artery.

but we observed an increased proportion of improved outcomes in patients with penumbral volumes <25 mL (Table 2). This likely reflects the tendency for distal

occlusions to present with smaller absolute penumbra volumes, rather than indicating a volume-specific treatment effect. A higher proportion of improved outcomes

Table 3. Primary Outcome (mRS Score, 0–1), Stratified by Occlusion Site and Perfusion Profiles

	Tenecteplase (n=242)	Alteplase (n=250)	Treatment effect* (95% CI)	<i>P</i> _{interaction} value
Occlusion site				
MCA (second segment [M2], proximal)	27/48 (56%)	29/50 (58%)	1.02 (0.74–1.41)	0.89
MCA (second segment [M2], distal)	30/46 (65%)	28/48 (58%)	1.14 (0.84–1.54)	
MCA (third segment [M3] and beyond distally)	62/81 (77%)	59/93 (63%)	1.23 (1.04–1.46)	
Anterior cerebral artery	12/21 (57%)	5/13 (38%)	1.46 (0.71–3.00)	
Posterior cerebral artery	21/46 (46%)	22/46 (47%)	0.92 (0.61–1.36)	
Ischemic core, mL†				
<10	99/151 (65%)	109/180 (60%)	1.04 (0.89–1.21)	0.69
10–20	38/61 (62%)	17/36 (45%)	1.43 (0.98–2.09)	
20–30	8/13 (61%)	12/19 (63%)	1.04 (0.64–1.70)	
≥30	7/17 (41%)	5/14 (35%)	1.37 (0.59–3.18)	
Core growth rate, mL/h‡				
<3	77/121 (64%)	84/142 (59%)	1.01 (0.85–1.21)	0.47
≥3	75/121 (62%)	59/108 (55%)	1.20 (0.98–1.48)	
Mismatch volume (penumbra), mL§				
<25	54/73 (74%)	55/101 (54%)	1.29 (1.04–1.59)	0.76
25–35	25/51 (49%)	31/51 (60%)	0.88 (0.63–1.22)	
35–45	37/55 (67%)	25/40 (62%)	1.12 (0.86–1.47)	
45–55	15/25 (60%)	12/23 (52%)	1.16 (0.71–1.89)	
≥55	21/38 (55%)	20/35 (57%)	0.95 (0.64–1.40)	

CTP indicates computed tomography perfusion; MCA, middle cerebral artery; mRS, modified Rankin Scale; and NIHSS, National Institutes of Health Stroke Scale.

*Adjusted risk ratio unless otherwise designated (adjusted for age, baseline NIHSS score, and baseline mRS score).

†Relative cerebral flow at a threshold of 30%.

‡Defined as core volume on baseline CTP (mL) divided by the time from stroke onset to initial CTP scan (hours).

§Defined as the difference between total perfusion volume and ischemic core lesion volume.

was observed among tenecteplase-treated patients with faster core growth rates ≥ 3 mL/h (Table 2; Tables S1 and S2). Although the underlying mechanism is uncertain, one plausible explanation is that the bolus administration of tenecteplase enables more rapid achievement of effective plasma concentrations,¹⁴ which may be advantageous in patients experiencing rapid core expansion. However, the lack of treatment interaction indicates that these results should be interpreted cautiously, and further work is required to validate these findings.

We made a conscious decision to exclude patients enrolled in TASTE with a large vessel occlusion, defined in this context as patients with an internal carotid artery, M1 segment, or vertebrobasilar artery occlusion. When TASTE was first launched in early 2014, the evidence supporting endovascular therapy had not been established. However, following the publication of the major landmark endovascular therapy trials,^{15–19} the number of patients with a large vessel occlusion enrolled in TASTE decreased significantly and, ultimately, made up <15% of the trial cohort. Given the extensive prior investigations of tenecteplase in the large vessel occlusion population,^{6,9,20–22} including in patients with no access to endovascular therapy,²³ we felt that the decision to focus on a more distal occlusion population was valid. The decision to include proximal M2 occlusions was made based on the evolving evidence regarding medium vessel occlusion stroke and the uncertain efficacy of endovascular therapy.^{24,25}

Our study has several limitations. First, while we observed a difference in clinical outcomes between tenecteplase and alteplase, we are not able to comment on rates of early recanalization, as this was not collected in the trial. Second, although the minimum mismatch volume and ratio criteria helped ensure a biologically coherent sample, the criteria also exclude patients with smaller lesions, which limits the generalizability of our findings. This was a post hoc secondary analysis, and CIs were not adjusted for multiple comparisons, increasing type I error. Therefore, it is important to acknowledge that the results are hypothesis-generating. Furthermore, interpretation of clinical outcomes among patients with ACA occlusions was constrained by the small sample size and wide CIs. Analysis of more distal PCA occlusions was also not possible due to the limited number of patients. A more comprehensive assessment of these populations may be more feasible in the future through individual-level data pooling across key comparative trials. It is also important to note that our analysis of ischemic core is limited to an assessment of reduced cerebral blood flow (<30%), yet there is growing evidence to show that markers of blood-brain-barrier breakdown on noncontrast CT are also predictive of poor outcome and postreperfusion complications.^{26–28} An assessment of these biomarkers within the TASTE cohort is planned in future analysis. Finally, while our analysis adjusted for age, baseline

NIHSS score, and prestroke mRS score aligned with the analytical methodology of the main trial, there are potential confounders, such as clot characteristics,²² which we did not adjust for in this present analysis.

To conclude, tenecteplase may offer a clinical advantage over alteplase for patients with distal vessel occlusions, particularly within the distal MCA territory. Symptomatic hemorrhage and mortality remain low across occlusion sites and perfusion profiles. Our hypothesis-generating results require further study to confirm our findings, but this analysis continues to reinforce the role of tenecteplase as a reliable agent within standard acute stroke workflows.

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APPENDIX

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